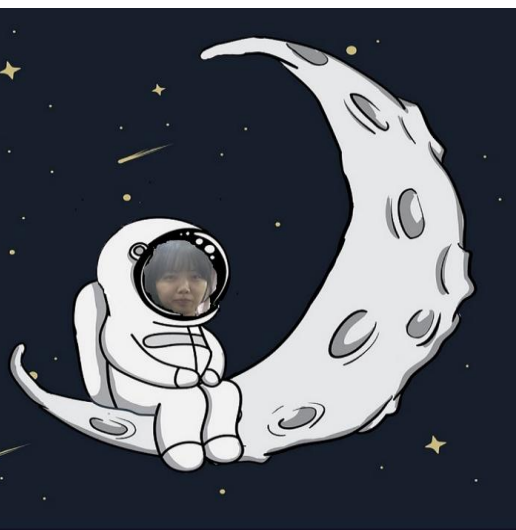


Name: _____ Class: _____ Student number: _____

1. (20 points) A particle of mass m in a Schwarzschild spacetime has an energy/mass of E and an angular momentum/mass of L . Its radial distance from the black hole is decreasing. For what values of E will the particle be captured by the black hole?

2. (33 points) Miss Zhang is in a circular orbit (which you can assume



is in the $\theta = \pi/2$ plane) around a spherical neutron star which does not rotate.

Her orbital radius is given by $r = 8GM$.

She is waiting on a planet there for the

crazy Lao Bi who has gone down to the

surface of the neutron star, $r = R_{NS} > 2GM$.

Lao Bi attempts to escape from the neutron star by launching his ship radially outward; however, the ship's speed is less than the escape velocity.

Miss Zhang sees Lao Bi pass her going outward.

Then Lao Bi reaches a maximum distance from the neutron star of H .

Then Lao Bi's ship falls radially inward again, again passing Miss Zhang, who has completed 12 of her orbits between these two encounters.

(a) How much time has elapsed on Miss Zhang's clock between encounters?

(b) If Lao Bi and his ship have a total mass m , what is the conserved total energy of Lao Bi & ship?

(c) Given your answer in (b), find an integral expression for the time elapsed on Lao Bi's clock between encounters.

The *definite* integral should be over the variable r and depend on H .

This is a rather ugly integral and you should **NOT** attempt to do it !!

(d) Has more, less, or the same time elapsed on Lao Bi's clock between the two encounters?

Please use physics arguments to justify your answer.



3. (12 points) Show that the Lorenz gauge in linearized gravity is equivalent

$$\text{to setting the condition: } g^{\mu\nu} \Gamma_{\mu\nu}^{\lambda} = 0 .$$

4. (35 points) Consider a radial ($d\Omega = 0$) null geodesic propagating in a closed FRW cosmology and the radial coordinate χ given by Eqn. (8.31) in Carroll. By appropriate normalization, χ may be taken as dimensionless and the positive curvature k (NOT dimensionful κ !!!) set to +1. Thus defined, a change in χ of 2π (radians) would correspond to going completely around the closed universe.

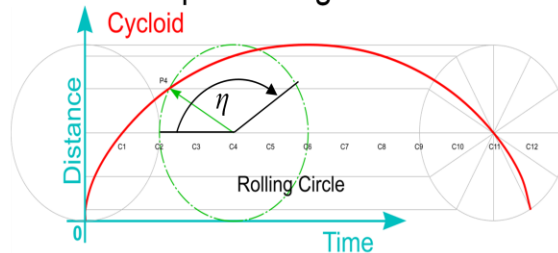
(a) Show that $\Delta\chi = \frac{1}{R_0} \int_{t_1}^{t_2} \frac{dt}{a(t)} .$

(b) Now consider a universe that is completely matter-dominated at all times (all other contributions are always negligible).

Show that the Friedman Equation allows for the following cycloidal parametrization:

$$a(\eta) = C (1 - \cos \eta) \quad t(\eta) = CR_0 (\eta - \sin \eta)$$

The variable η is called the development angle. The situation is shown pictorially here:



Determine the constant C in terms of the current energy density of the universe, ρ_0 , and R_0 .

(c) At $\eta = 0$, $t = 0$ and $a = 0$. This is the Big Bang. If $a = 0$ at a later time, then that would be the Big Crunch. How far around the universe would a light ray emitted at the time of the Big Bang be able to travel by the time of the Big Crunch?

(d) Next consider a universe with only radiation. Now the cycloidal parametrization is no longer valid. Show that instead $a(t)$ may be written as

$$a(t) = B \sqrt{1 - [1 - t/(R_0 B)]^2}$$

Determine the constant B in terms of the current energy density of the universe, ρ_0 , and R_0 .

(e) Now how far around this universe would a light ray emitted at the time of the Big Bang be able to travel by the time of the Big Crunch? *Hint:* the variable substitution $w = 1 - t/(R_0 B)$ should prove helpful.